
Web and Multimedia Technologies

Sample exam paper 2

Notes:

- The exam paper is printed double-sided on two A3 sheets
- The exam duration is 2 hours and 45 minutes
- The maximum grade achievable in the written test is 30

Università di Pavia
Master's Degree in *Computer Engineering*

Web and Multimedia Technologies exam (prof. Marco Porta)

Date: *Month day, year*

Surname _____ Name _____

Student ID _____

Always justify your answers. Fold this sheet in half with this side as the cover. If you require additional space, use the draft (which must be submitted) and any extra sheets (available on request). Ensure that your details are written on every sheet.

1. Say whether the following statements are true (T) or false (F), clearly justifying your choices (3.5)

a. In the context of XML, *DTDs* and *XML Schemas* have the same function

☐ T ☐ F

b. CSS *absolute positioning* always arranges elements with respect to the browser's window

☐ T ☐ F

c. SVG is a language for the description of vector graphics

☐ T ☐ F

d. The *RGB* color model is more suitable for *warm* colors, while the *CMYK* model is more suitable for *cool* colors

☐ T ☐ F

e. The *color lookup tables* of any two GIF images always have at least one color in common

☐ T ☐ F

f. *DHTML* is a new HTML with more tags than HTML5

☐ T ☐ F

g. *Multimedia* interfaces are always *multimodal* interfaces

☐ T ☐ F

2. What is the purpose of the *Color Management System* (CMS) in software tools like GIMP or Adobe Photoshop? How does it work? (2)

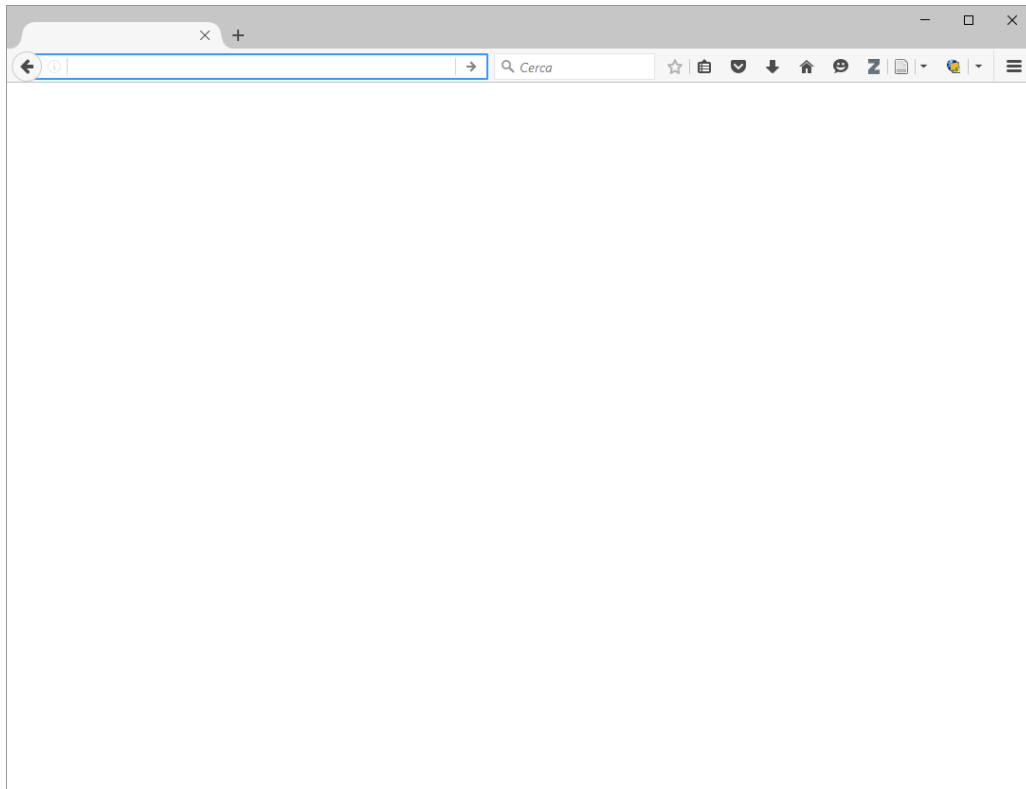
3. Given the following HTML/CSS code, provide a qualitative graphic representation of the resulting web page (5)

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="utf-8">
    <title>University of Pavia</title>
    <style>
      body, h1 { margin: 0px; padding: 0px }
      #content {
        padding: 0px; margin: 0px;
      }
      #menu {
        float: right; background-color: yellow;
      }
      #footer {
        background-color: lightgrey;
        height: 30px; text-align: center;
      }
      td {padding: 10px}
    </style>
  </head>
  <body>
    <nav id="menu">
      <ul>
        <li><a href="syllabi/ee.html">Electronic Engineering</a></li>
        <li><a href="syllabi/ce.html">Computer Engineering</a></li>
        <li><a href="syllabi/iae.html">Industrial Automation Engineering</a></li>
      </ul>
    </nav>
    <article id="content">
      <table style="border: solid #ff0000 1px; width: 70%; margin-left: auto; margin-right: auto">
        <tr>
          <td>
            
          </td>
          <td rowspan="2">
            The School of Engineering of the University of Pavia (Italy) offers three international
            Master of Science programs ("Lauree Magistrali"): MSc in <em>Electronic Engineering</em>,
            MSc in <em>Computer Engineering</em>, and MSc in <em>Industrial Automation Engineering</em>.
          </td>
        </tr>
        <tr>
          <td>
            The University of Pavia is one of the oldest in Europe, founded in 825.
          </td>
        </tr>
      </table>
    </article>
    <div id="footer">&copy; Copyright 2026</div>
  </body>
</html>
```

Also, answer the following questions:

- With respect to the directory containing the above-defined HTML file, where is the file `ee.html`?
- Supposing that the administrator of the web server has set `c:/myroot` as the root directory, what is the path (i.e., the actual position) of the file `unipv.gif` on the server? (`c:/...`)
- Is `<p style="font-weight: bold">` semantically equivalent to `<strong>`?

a.	
b.	
c.	



4. Define an external style sheet named "mystyle.css" which specifies that: (1) the page has a 25-pixel margin on all four sides and a black background; (2) the class *ita* for the `<td>` element is characterized by italic text (value *italic* for the *font-style* property). Indicate (clearly): (a) the content of the CSS; and (b) the HTML code necessary to [1] load the external CSS file and to [2] make a `<td>` tag apply the *ita* class (2)

CSS content	HTML code
	1
	2

5. Showing the operations (it is not necessary to perform the calculations), indicate: (1)

a.	The size of the file resulting from the (uncompressed) digitization of 5 seconds of audio signal (stereo, 15 kHz, 16 bits)
b.	The width and height (in pixels) of the image obtained from the scan of a photograph, 5 inches wide and 7 inches high, with a 600-dpi resolution
c.	The size of the file resulting from the (uncompressed) digitization of 3 minutes of video signal (640x480 pixels, 24 bits, 25 fps)

6. Explain the following assertion: "The *RGB* and *CMY* color models are complementary" (2)

7. In an area of a web page, there is the photo of a cat that moves from left to right and vice versa; when the photo (not a region external to it) is clicked, another page is loaded. Say (explaining why) which of the following technologies can be exploited to achieve such a behavior, and which ones cannot (2)

HTML alone ☐ Yes ☐ No

CSS ☐ Yes ☐ No

JavaScript ☐ Yes ☐ No

Animated GIFs ☐ Yes ☐ No

CGI programs or application servers ☐ Yes ☐ No

8. How does the audio digitization process occur? What parameters must be taken into account? Can the *MP3* format be considered somewhat similar to the *MIDI* format? Why? (2)

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Fold this sheet in half with this side facing up, and insert it into Sheet 1 before submission.

9. Given the following HTML/JavaScript code, concisely (but clearly) describe the meaning of each line marked with a number. Then, also explicate the "general behavior" of the page (3)

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="utf-8">
    <title>JavaScript test</title>
    <script>
      var cur_im = 0;
      var img = new Array(3);
      var addr = new Array(3);
      img[0] = "google.png"; img[1] = "bing.png"; img[2] = "yahoo.png";
      addr[0] = "https://www.google.com";
      addr[1] = "https://www.bing.com";
      addr[2] = "https://www.yahoo.com";

      function n() {
        cur_im = cur_im + 1;
        if ( cur_im == 3 )
          cur_im = 0;
1.    document.getElementById("im").src = img[cur_im];
      }
      function loadP() {
2.    document.location.href = addr[cur_im];
      }
    </script>
  </head>
  <body>
3.    <a href="javascript:loadP()" ">
      
    </a>
    <form name="f">
4.    <input type="button" value="Change" onclick="n()" ">
    </form>
  </body>
</html>
```

1.	
2.	
3.	

4.

General page behavior

10. Supposing that the form defined at the previous point 9 also includes a text field and a button, write the HTML and JavaScript code necessary to obtain the following behavior: if a color is entered in the text field (e.g., *yellow* or *#f528d7*), then, when the button is pressed, the background of the page is changed to that color; if, instead, the text field is empty when the button is pressed, an error message is displayed. Suppose that: (1) if something is entered in the text field, then it is certainly a color (no check is necessary); (2) an `exists` function is already available which returns `true` if its input parameter contains at least one character, `false` otherwise (2.5)

11. What is XSLT? Is it necessarily connected with the concept of a *web browser*? Provide at least one example of its use (2)

12. The entity *Apartment* is informally described as follows:

An apartment is characterized by its address, floor, total size (in square meters), room sizes, and price (in euros).

To formally describe the entity using an XML language, define proper tags and attributes, and then use them to represent the *apartment located in Milan, at Via Verdi n. 10, 4th floor, 100 m² in total, priced at 450,000 euros, featuring:*

- *A kitchen (25 m²)*
- *A living room (20 m²)*
- *Two bedrooms (20 m² and 23 m²)*
- *A bathroom (12 m²)*

Apply the tags and attributes directly to the example, and then define the DTD.

Important: use at least one attribute (3)

